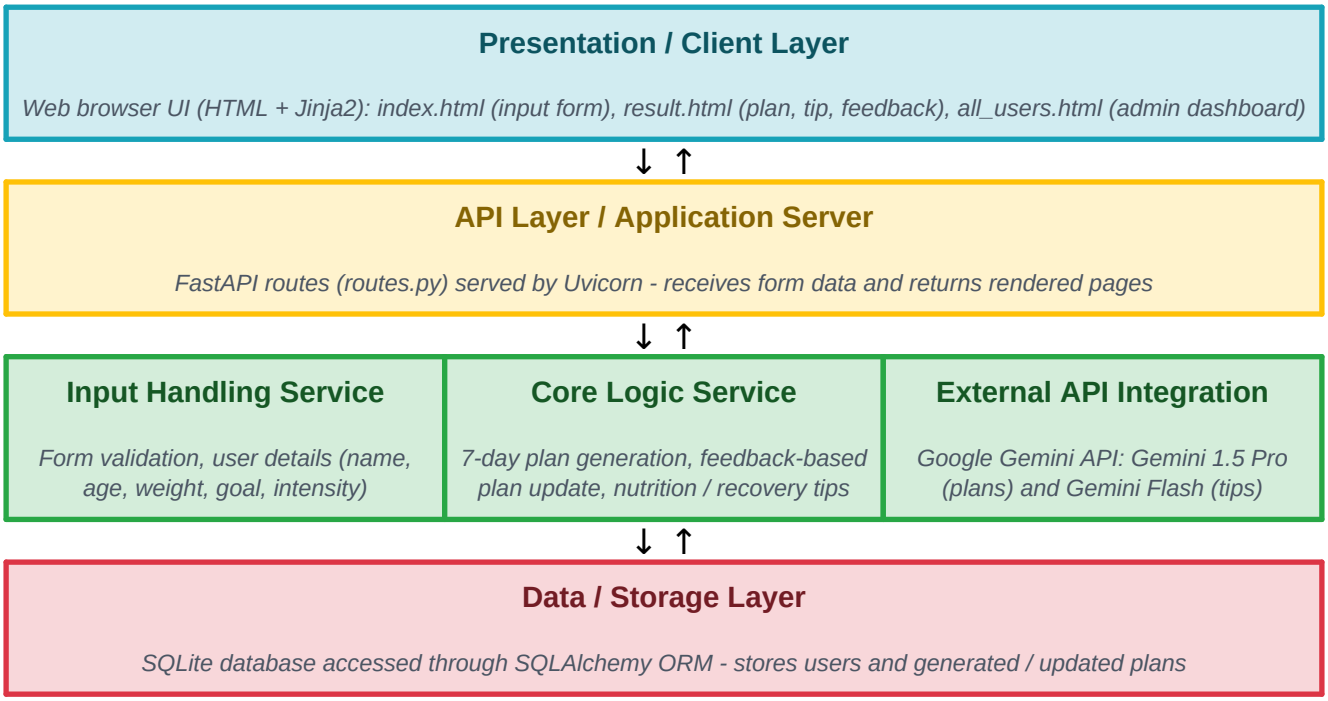
Solution Architecture

Date	15 March 2026
Team ID	unm293200909
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Name	Kavyasri.A
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of FitBuddy's Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used	Work Assigned to Kavyasri.A
Presentation Layer	Collects user details and shows the 7-day plan, nutrition tip, feedback form and the admin user list.	HTML, CSS, Jinja2 templates	Design index.html, result.html and all_users.html
API Layer	Receives requests, routes them to the right logic and returns rendered pages or API responses.	FastAPI, Uvicorn, routes.py	Build FastAPI app and routes
Core Logic Service	Builds prompts, generates the 7-day plan, updates it from feedback and produces goal-based tips.	Python	Write plan, update and tip logic
External API (Gemini)	Generates workout plans and updates with Gemini 1.5 Pro; generates quick tips with Gemini Flash.	Google Gemini API (google-generativeai)	Set up API key and prompts
Database	Stores user details, original and updated plans so they can be viewed later by users and admin.	SQLite, SQLAlchemy ORM	Create models and database handling